

Peer Review: Investigating Algorithmic Bias’ Effect on Social Media Content Curation

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Summary

This midterm report investigates algorithmic bias in AI-powered social media content moderation, focusing on “shadowbanning” during the 2024 U.S. election. Using the USC X 2024 Election dataset, the authors embed and cluster tweets into topics and analyze a key metric—views/likes—to detect content suppression. The study compares topic distributions against Gaussian baselines and employs statistical tests (e.g., t-tests) to examine potential bias. Although the final discussion is not yet complete, the paper outlines a clear and structured approach to a timely issue.

Strengths

1. **Relevant and Timely:** The topic of algorithmic bias, especially during elections, is critical. It contributes meaningfully to ongoing discussions about fairness and transparency in AI systems.
2. **Thoughtful Methodology:** The approach combines clustering, topic modeling, and statistical inference to quantify suppression. The novel use of the views/likes ratio as a proxy for shadowbanning is a creative metric for engagement analysis.
3. **Engagement with Literature:** The paper draws from a well-rounded set of prior studies, situating itself within current research and building on existing findings.

Areas for Improvement

1. **Incomplete Analysis:** Only one week of data is analyzed. Adding preliminary insights, even on limited data, would help validate the methodology and strengthen the narrative.
2. **Unclear Technical Details:** The clustering and embedding methods are not clearly specified. Identifying models (e.g., BERT, KMeans) would improve reproducibility.
3. **Definition of Shadowbanning:** More discussion is needed on why the views/likes ratio is chosen. Alternative explanations (e.g., time of day, follower count) should be considered, or controlled for.

Other Comments

Minor issues include placeholder text in the abstract and underexplained visuals (e.g., Figure 1). Future versions should ensure all figures are clearly described. A brief reflection on the ethics of demographic inference would also enrich the analysis.